MAS508-G Control Theory
Lecture 2: System Modeling

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Lecture 2d. Experimental Modeling

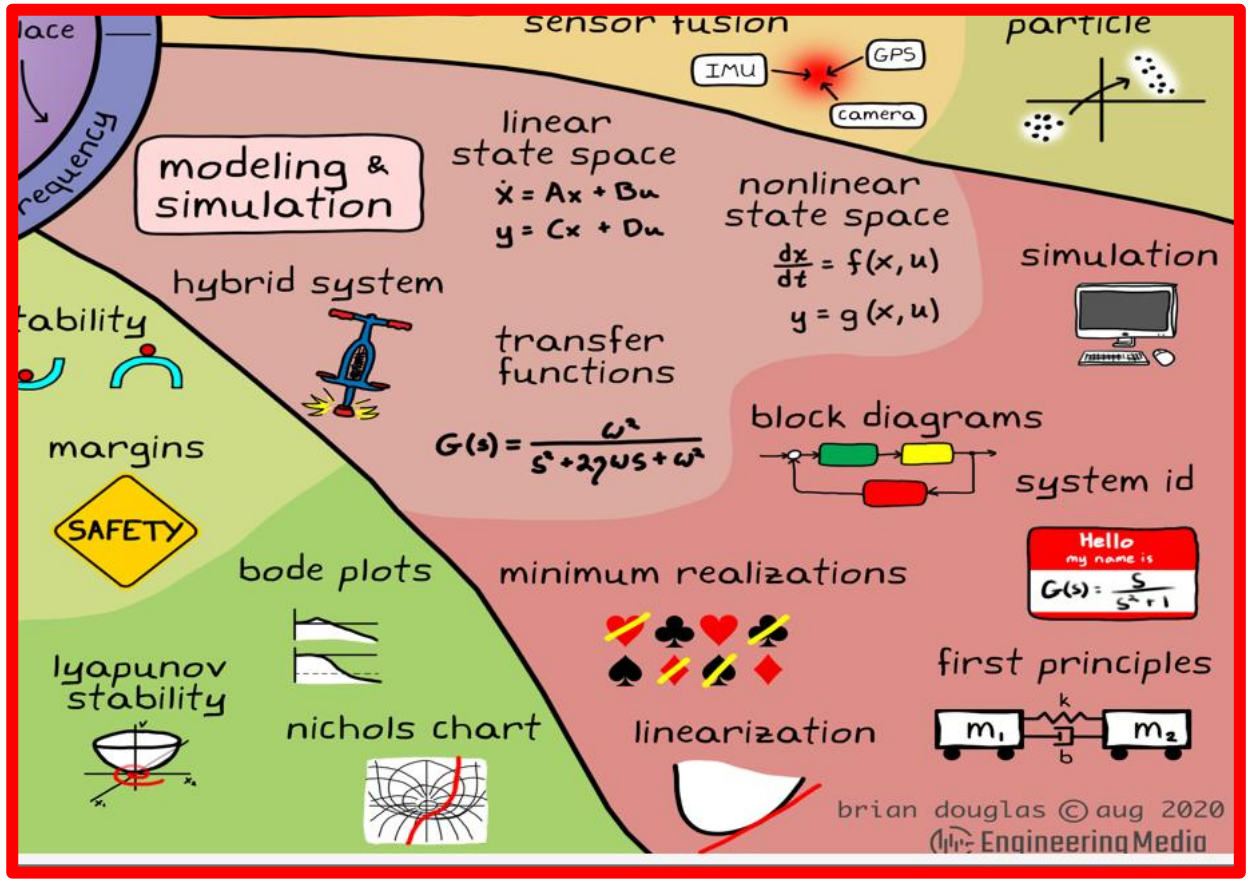
Lecture 2d Experimental Modeling

a) Review of Theoretical Modeling
   - Differential equations
   - Transfer functions
   - Block diagrams
   - Frequency response

b) State-Space Representation

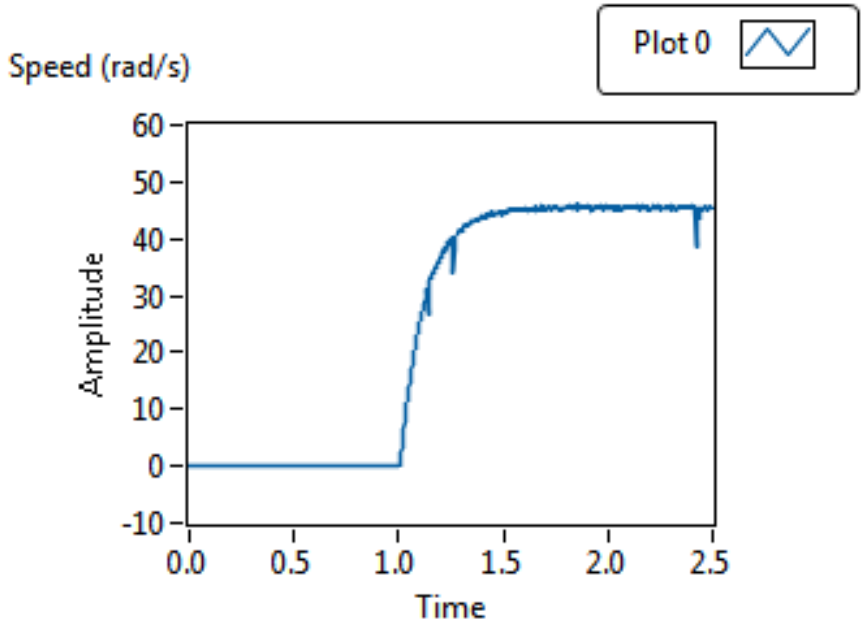
c) Linearization

d) Experimental Modeling



What is *experimental* model?

- Mathematical model is derived by examining a system's response to different inputs
- Also known as "black box" modelling
  - Don't need to know the details of the system (e.g. dynamics, system parameter)



Bump test method applies a step input. Model is derived from output response (in this case, motor speed).

Why derive a model *experimentally*?

1. Sometimes it's **necessary**.
   - You don't have the system specifications.
2. Can be more **accurate**
   - Modelling something experimentally can capture dynamics that may otherwise be forgotten when deriving it analytically
3. Easier
   - Modelling system theoretically can be quite tedious, especially in complex ones

Experimental modelling

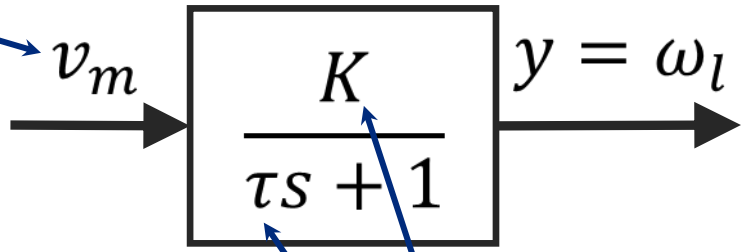
- Time domain: Step response or Impulse response
  - 1st-oder system
  - 2nd-order system

- Frequency domain: sinusoid response, Bode plots

Experimental Modelling of First-Order Systems

1) Apply step voltage

2) Measure corresponding speed



3) Find model parameters from speed response

1st-oder Transfer function

$$G(s) = \frac{K}{\tau s + 1}$$

- K is called the "gain"
- τ is called the "time constant"
- Unit step response: $K\left(1 - e^{-t/\tau}\right)$

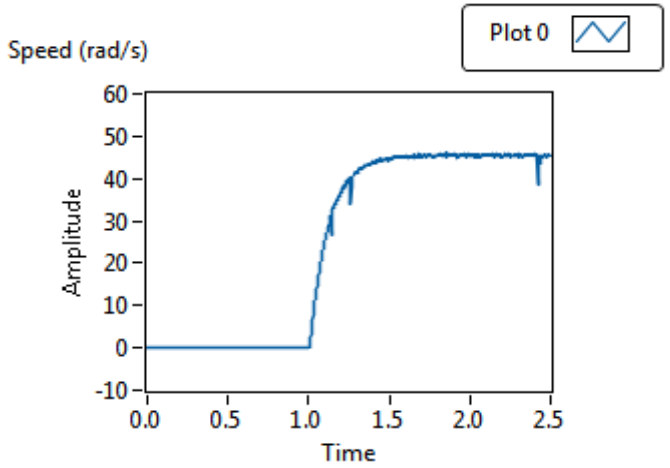


Fig. Step response of motor speed

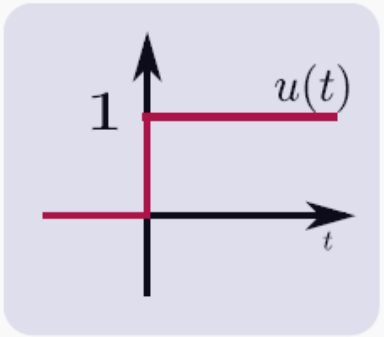
First-Order Systems

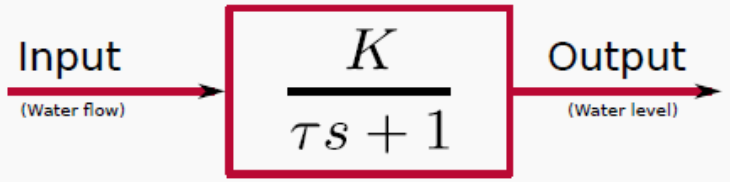
1st-oder Transfer function

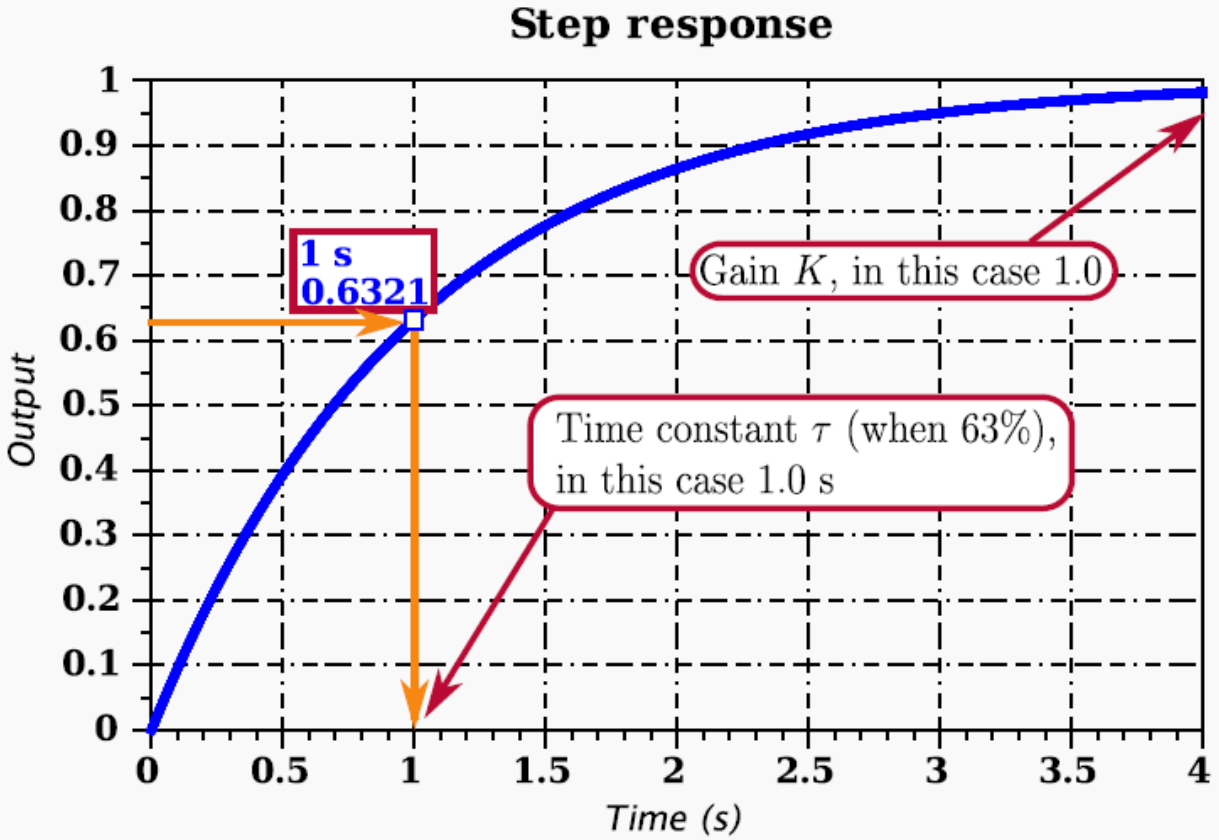
$$G(s) = \frac{K}{\tau s + 1}$$

- K is called the "gain"
- τ is called the "time constant"
- Unit step response: $K\left(1 - e^{-t/\tau}\right)$

- Time Constant τ : Indication of speed of response (convergence) Response is faster as τ becomes smaller.
- Time Constant τ : is the time period in which the step response reaches **63%** of its final value.





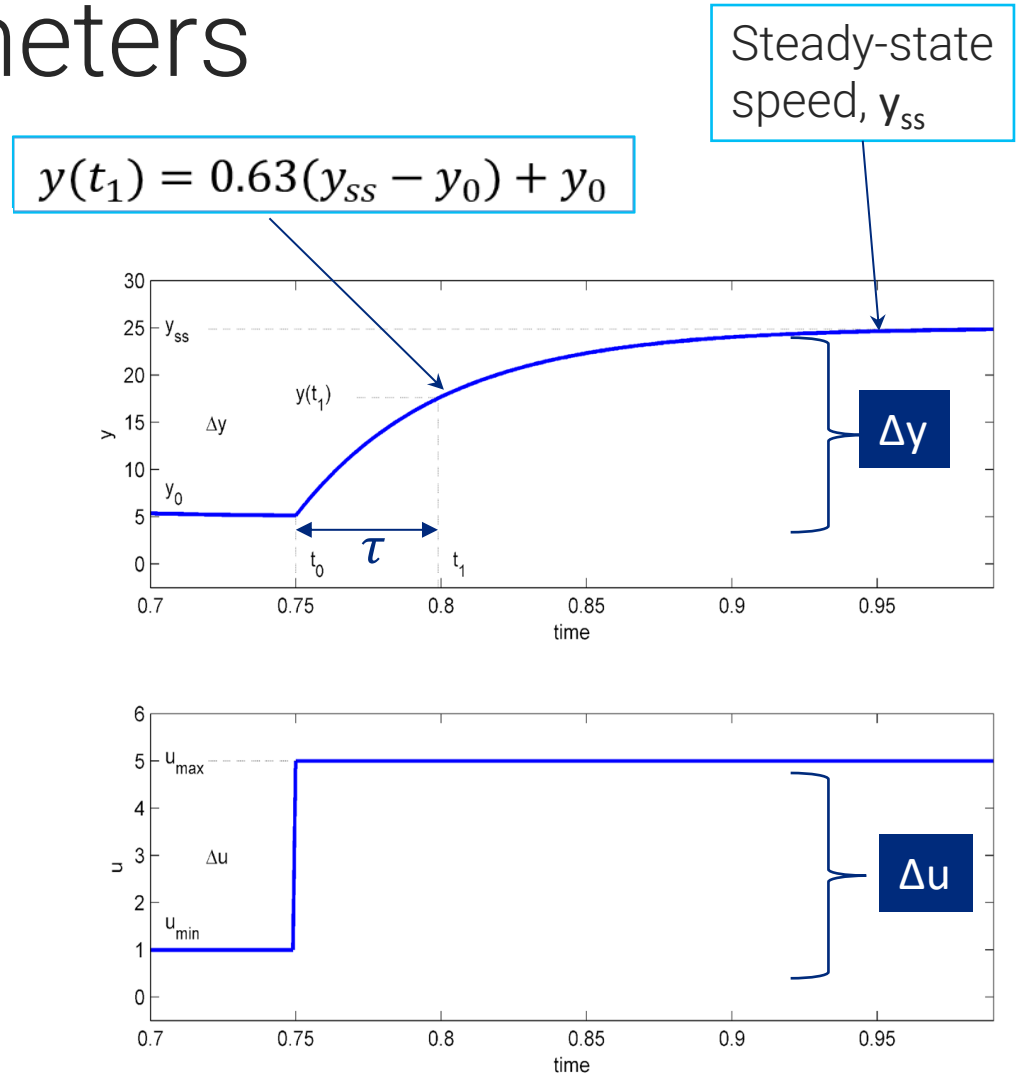


Finding 1st Order Parameters

- Find *K* and τ using step response measurements
- Measure values from speed step response
  - Steady-state speed, y_{ss}
  - 63% decay:

$$y(t_1) = 0.63(y_{ss} - y_0) + y_0$$

  - $K = \Delta y / \Delta u$
  - $\tau = t_1 - t_0$

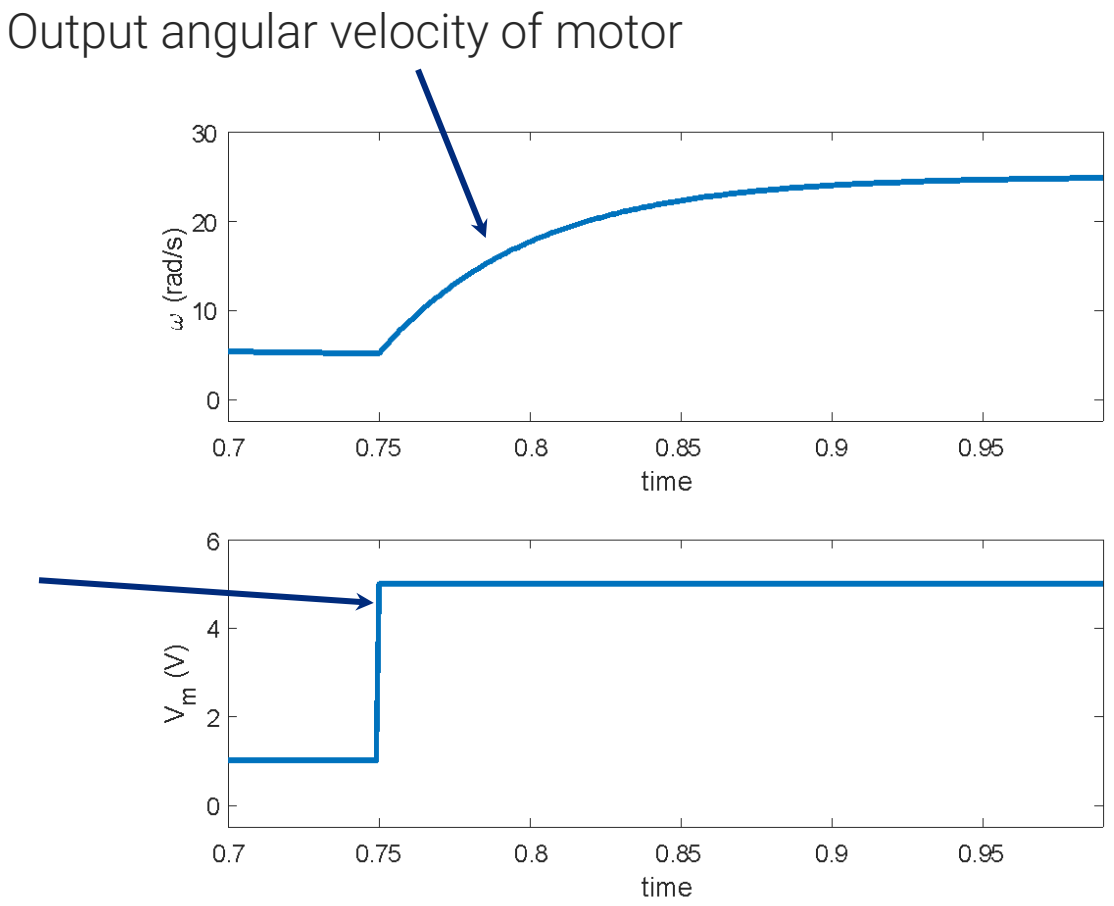


Example: DC Motor

Find the first-order transfer function model parameters for a DC motor:

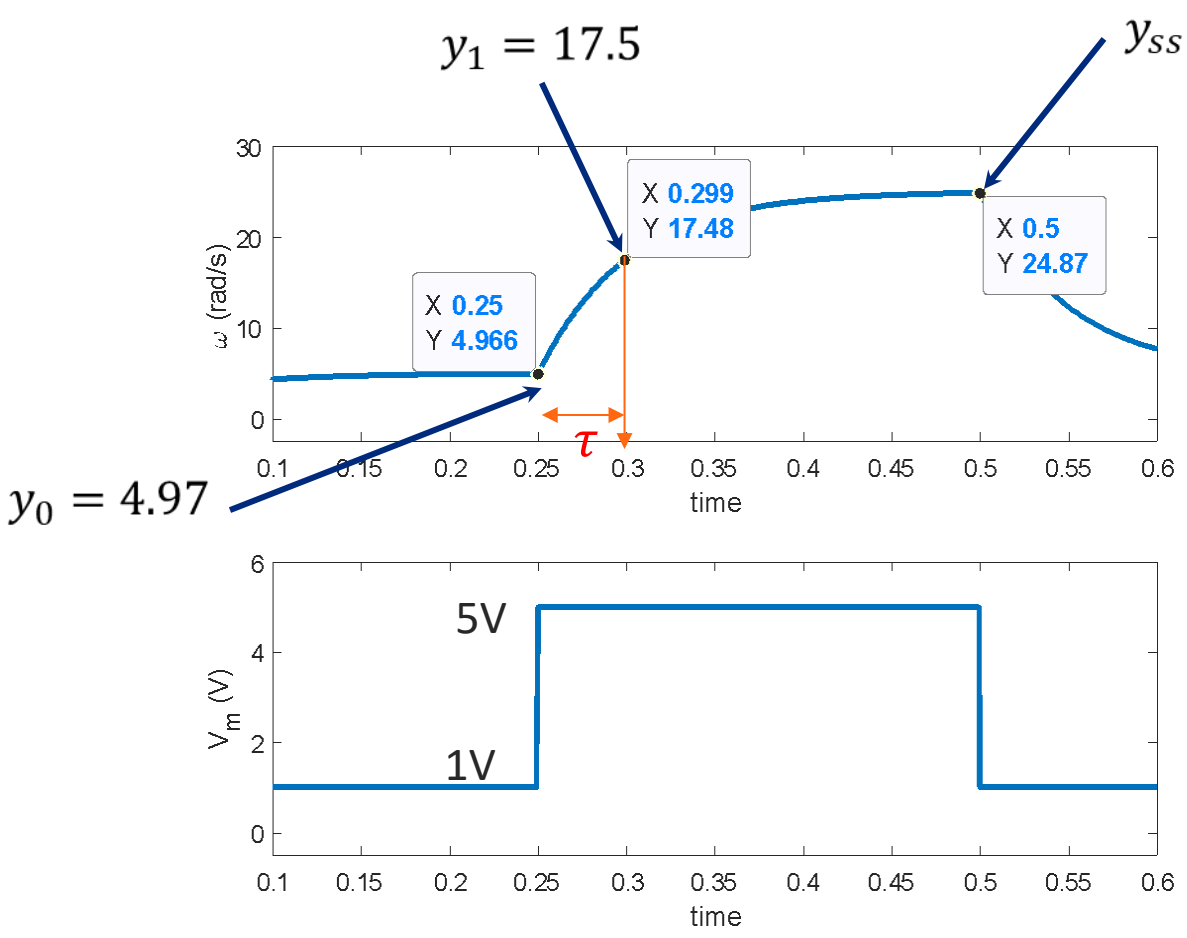
$$\frac{\Omega(s)}{V_m(s)} = \frac{K}{\tau s + 1}$$

- Apply a step input
- Examine the output response
- Determine 1st-order system
- Finding the transfer function



Step response of DC motor hardware

Example: DC Motor



(1) Find steady-state gain: $K = \frac{\Delta y}{\Delta u} = \frac{y_{ss} - y_0}{u_{max} - u_{min}}$

$$K = \frac{24.87 - 4.97}{5 - 1} = 4.98$$

(2) Find time constant: $y(t_1) = 0.63(y_{ss} - y_0) + y_0$

$\tau = t_1 - t_0$

$$y_1 = y(t_1) = 0.63(24.87 - 4.97) + 4.97 = 17.5$$

$$\tau = 0.30 - 0.25 = 0.05$$

Second-Order Systems

General 2nd Order Transfer Function

$$\frac{C(s)}{R(s)} = G(s) = \frac{K}{\frac{s^2}{\omega_n^2} + \frac{2\zeta}{\omega_n}s + 1}$$

$$= \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- K - is the "gain"
- ω_n - is the "natural frequency"
- ζ - is the "damping ratio"

ω_n is the frequency of oscillation of the system without damping

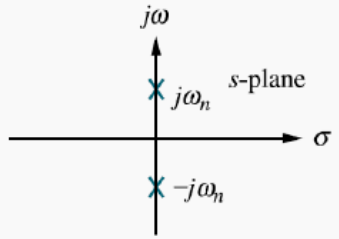
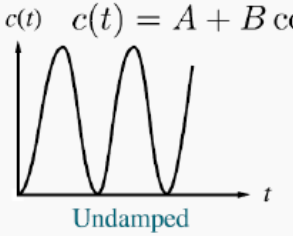
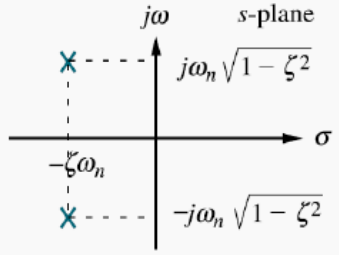
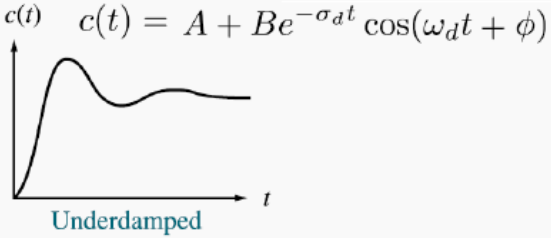
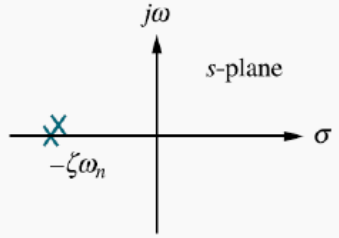
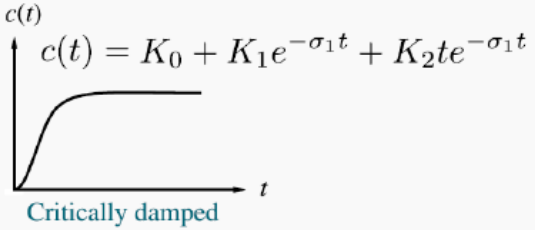
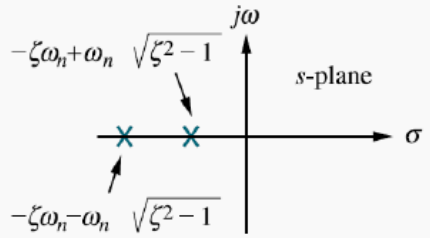
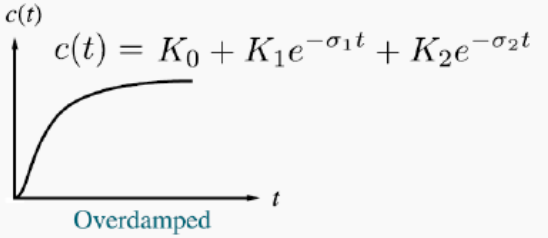
$$\zeta = \frac{\text{Exponential decay frequency(rad/sec)}}{\text{Natural frequency(rad/sec)}}$$

Second-Order Responses

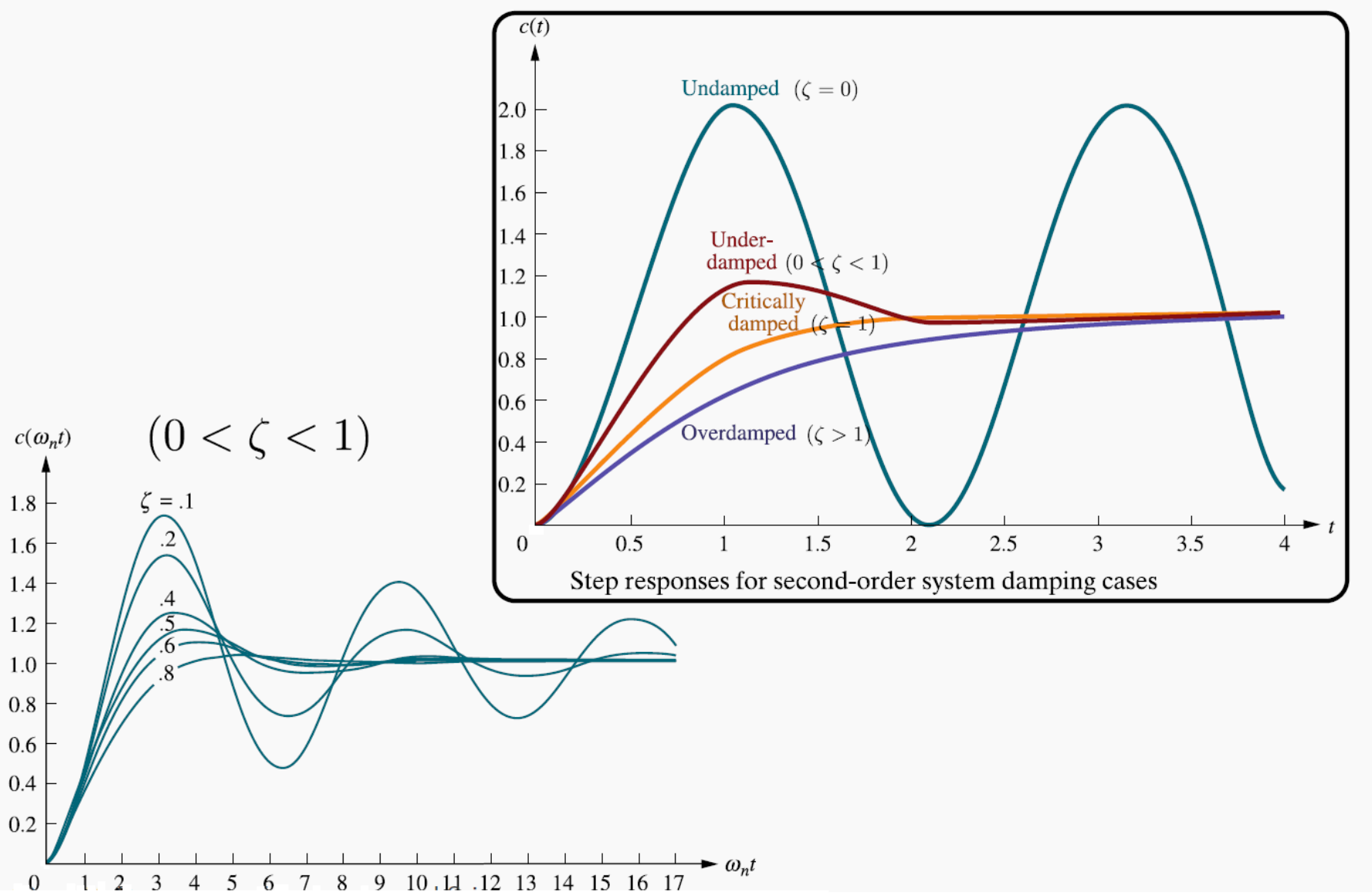
$$G = \frac{K}{\frac{s^2}{\omega_n^2} + \frac{2\zeta}{\omega_n}s + 1}$$

Roots:

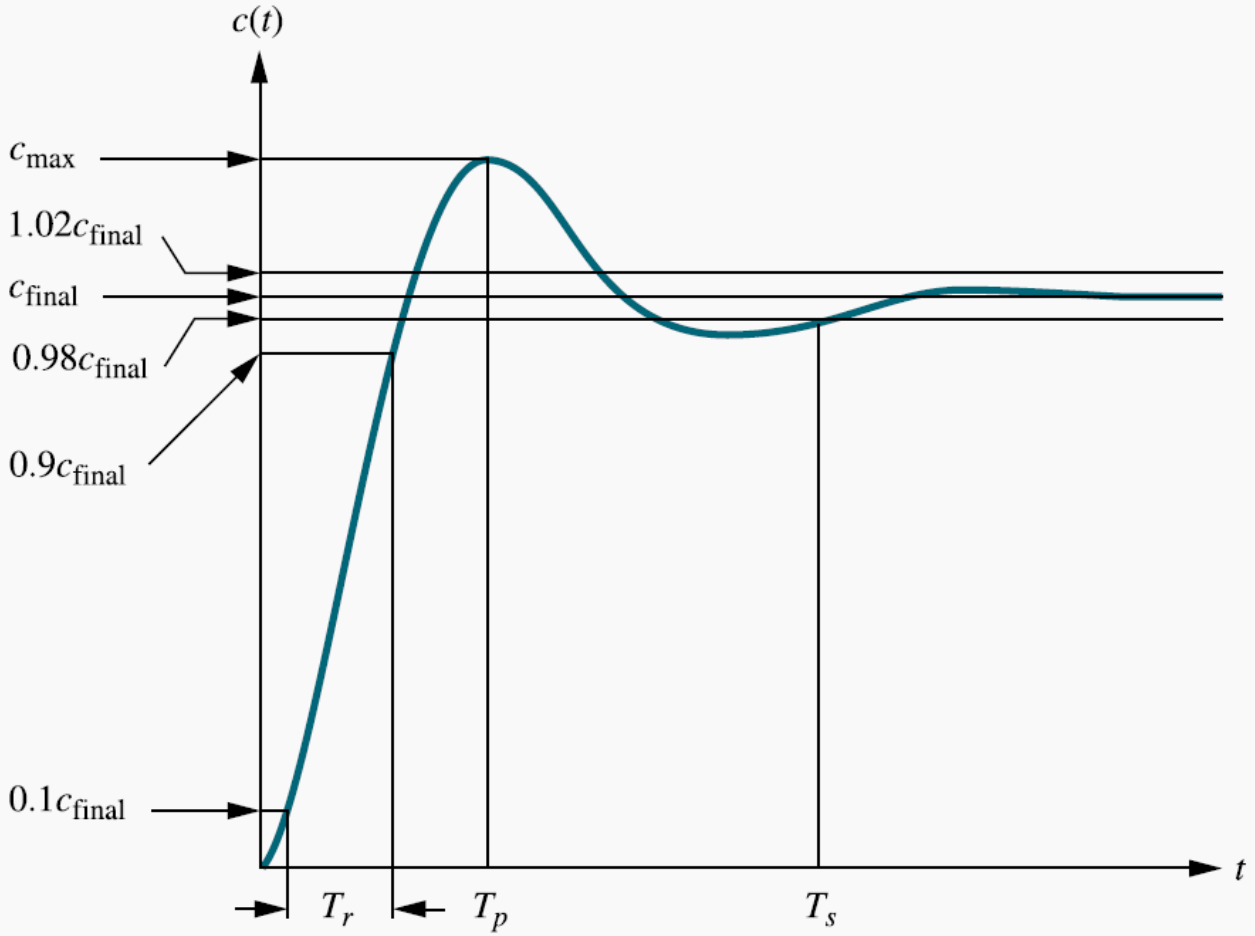
$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

ζ	Poles	Position	Step response
0		Two imaginary at $\pm j\omega_1$	
$0 < \zeta < 1$		Two complex at $-\sigma_d \pm j\omega_d$	
$\zeta = 1$		Two real at $-\sigma_1$	
$\zeta > 1$		Two real at $-\sigma_1$, $-\sigma_2$	

Second-Order Responses



Second-Order Under-Damped Systems



Second-order underdamped response specifications

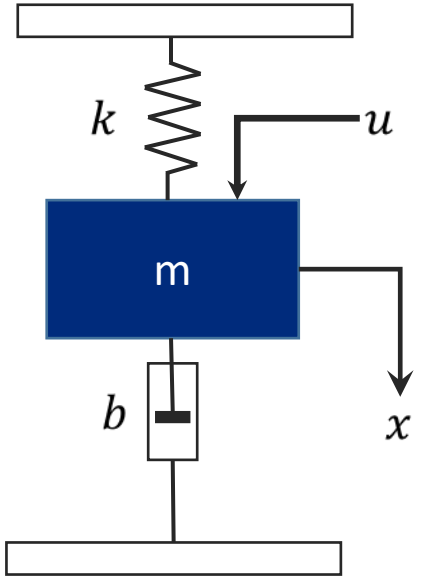
Rise time: T_r
Time to go from 10% of the final value to 90% of the final value.

Peak time: T_p
Time to reach the first, or maximum peak.

Percent overshoot: $\%OS$
Amount that the waveform overshoots the steady-state value at the peak time. (percentage of the final value: $\frac{c_{max}-c_{final}}{c_{final}}$)

Settling time 2%: T_s
Time to reach and stay within $\pm 2\%$ of the steady-state value.

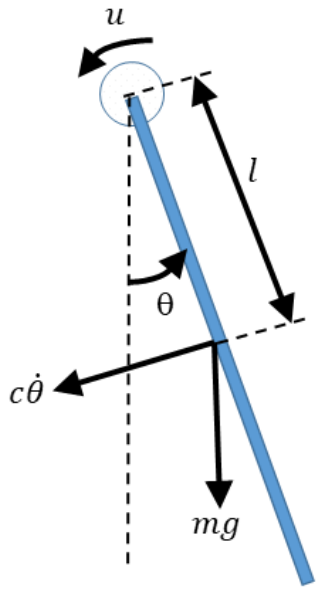
Modelling Second-Order Systems



Linear system with mass (m), viscous damping (b), and stiffness (k) with an input force (f_c):

$$m\ddot{x} + b\dot{x} + kx = u$$

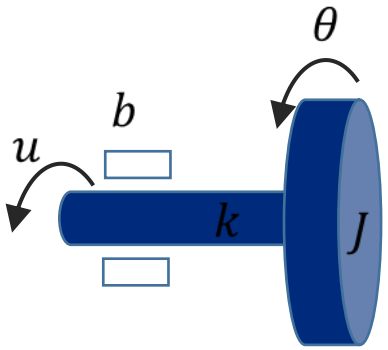
$$\frac{X(s)}{U(s)} = \frac{1}{ms^2 + bs + k}$$



Pendulum with mass (m), viscous damping (c), and stiffness (k) with an input torque (u):

$$ml^2\ddot{\theta} + c\dot{\theta} + mgl\sin\theta = u$$

$$\frac{\Theta(s)}{U(s)} = \frac{1}{ml^2s^2 + cs + mgl}$$



Torsional system with inertia (J), viscous damping (b), and shaft stiffness (k) with an input torque (u):

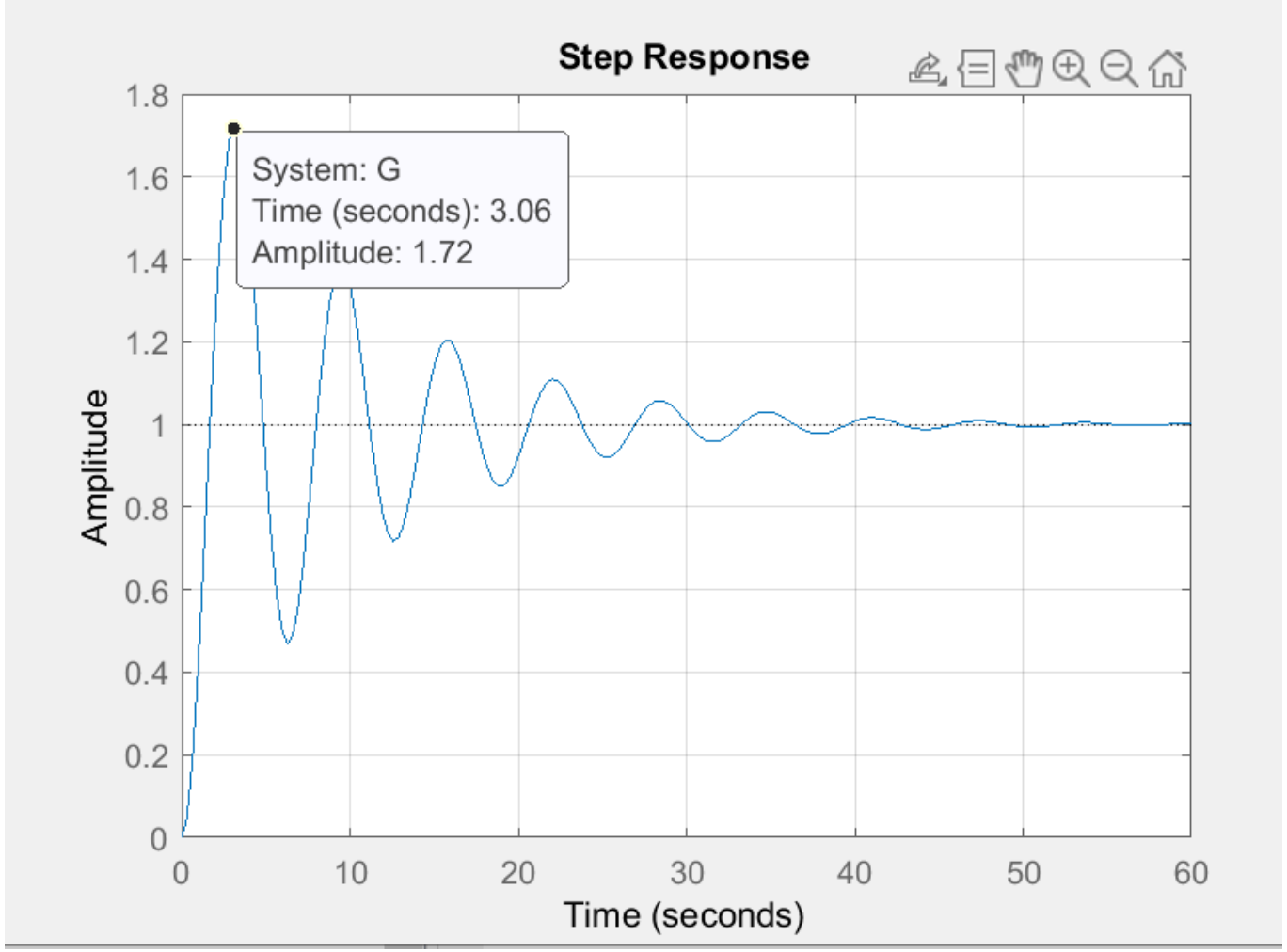
$$J\ddot{\theta} + b\dot{\theta} + k\theta = u$$

$$\frac{\Theta(s)}{U(s)} = \frac{1}{Js^2 + bs + k}$$

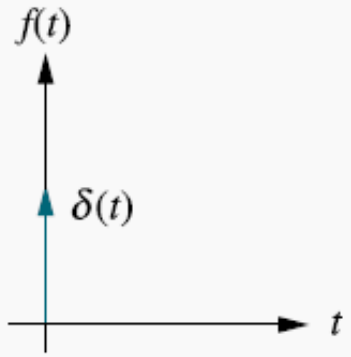
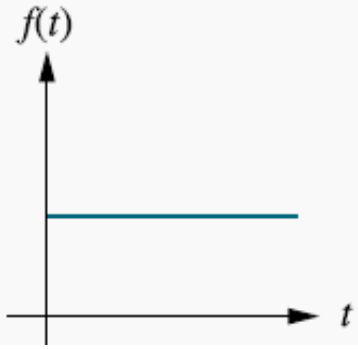
Finding System Model

- System parameters can be found from the response
- To do this experimentally, apply a *step input* and *examine the output response*

1) Step response
2) Impulse response

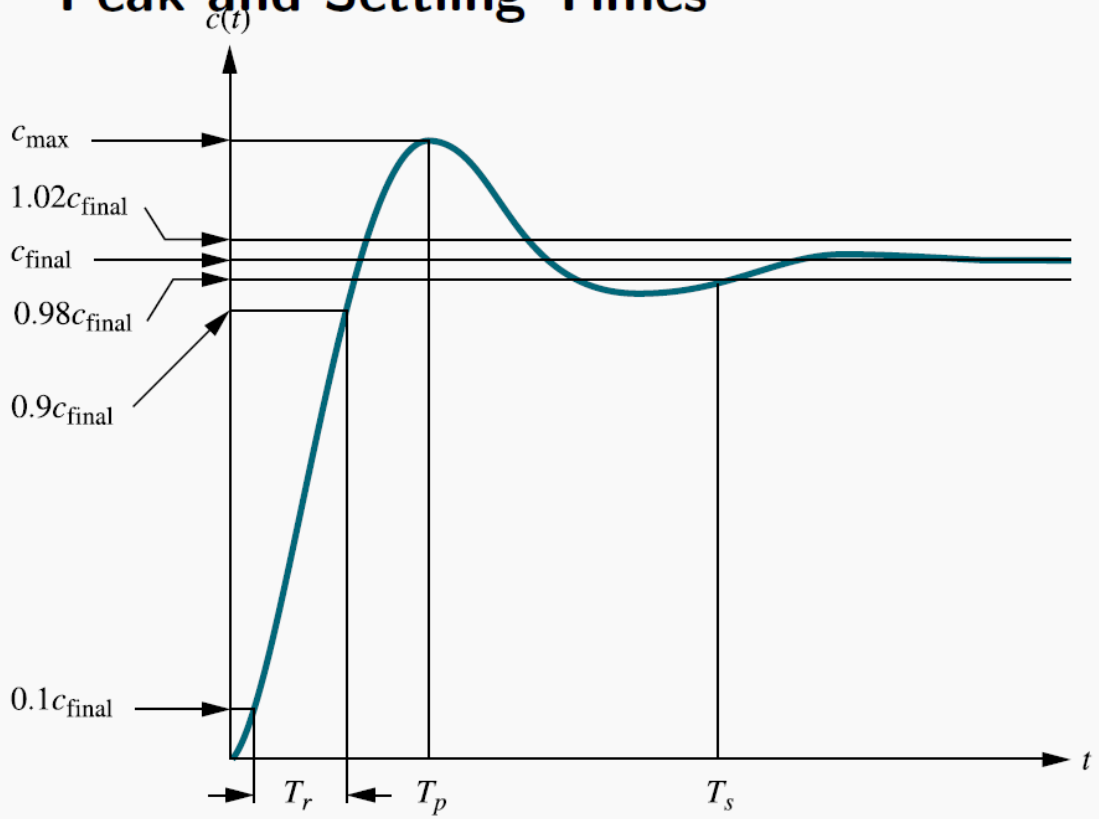


Input signals

Input	Function	Description	Sketch	Use
Impulse	$\delta(t)$	$\delta(t) = \infty$ for $0- < t < 0+$ $= 0$ elsewhere $\int_{0-}^{0+} \delta(t)dt = 1$		Transient response Modeling
Step	$u(t)$	$u(t) = 1$ for $t > 0$ $= 0$ for $t < 0$		Transient response Steady-state error

Second-Order Systems – Step Response

Peak and Settling Times



Second-order underdamped response specifications

$$OS = \frac{c_{max} - c_{final}}{c_{final}}$$

$$\%OS = OS \times 100$$

Natural frequency

!!! $\ln = log_e \neq \log_{10}$!!

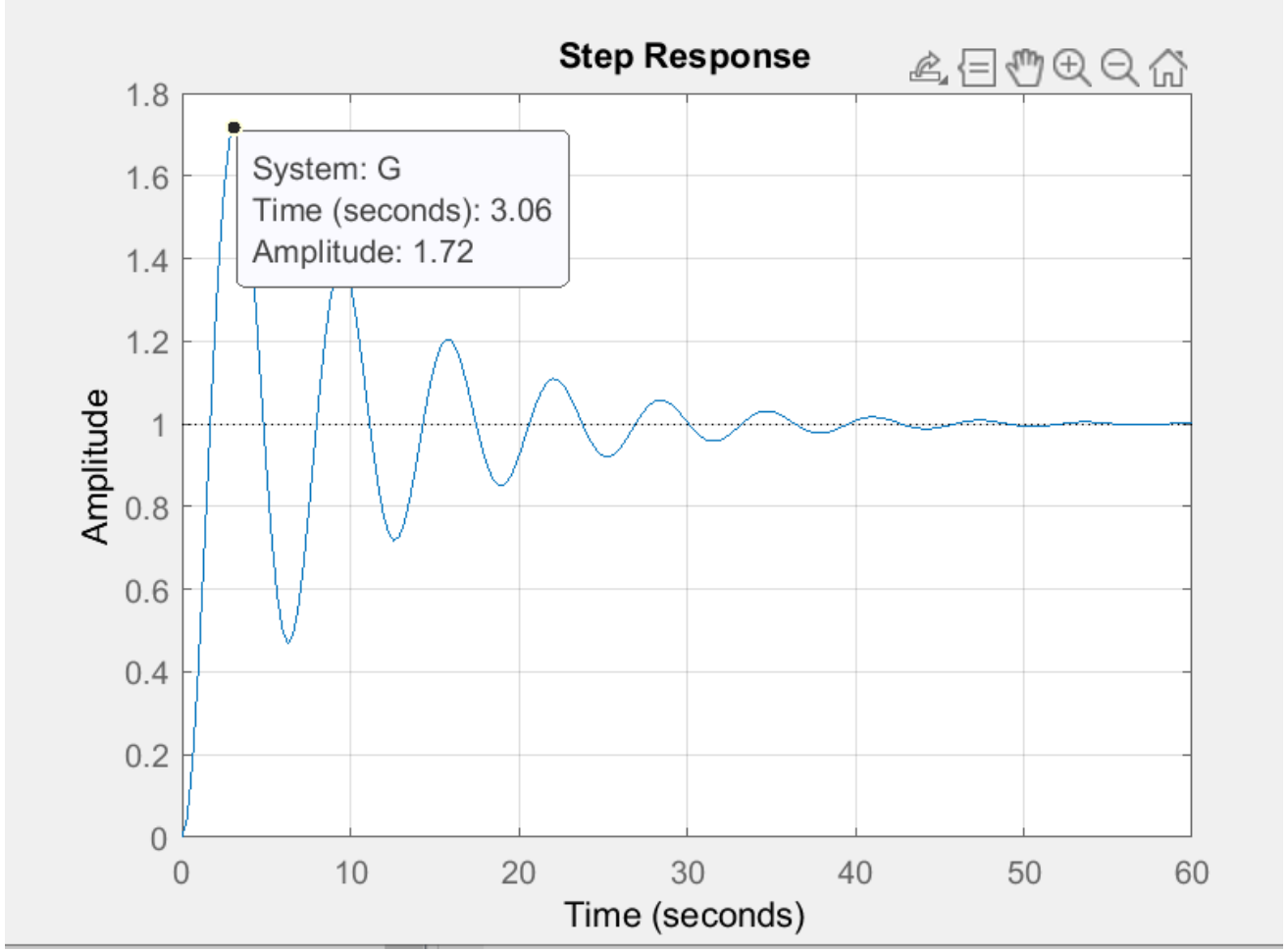
$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$

$$T_s = \frac{4}{\zeta \omega_n}$$

$$\%OS = e^{-\left(\zeta\pi/\sqrt{1-\zeta^2}\right)} \times 100$$

$$\zeta = \frac{-\ln(OS)}{\sqrt{\pi^2 + \ln^2(OS)}}$$

1) Step response: Finding Natural Frequency and Damping Ratio



Find the gain $\mathrm{K} = \frac{Final\ value\ of\ output}{Amplitude(step)} = \frac{1}{1} = 1$

Finding the damping ratio, ζ

$$\%\mathrm{Os}/100 = \frac{\max - \mathrm{finial}}{\mathrm{final}} = \frac{1.72 - 1}{1} = 0.72$$

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}} \approx 0.1$$

Finding the natural frequency, ω_n

$$T_p = 3.06$$

$$\omega_n = \frac{\pi}{T_p\sqrt{1-\zeta^2}} \approx 1.0$$

mass m=1 kg.

Finding Model Parameters

Assuming a mass-spring-damper system:

$$m\,\ddot{x}(t) + b\,\dot{x}(t) + kx(t) = \mathrm{u(t)}$$

Transfer function:

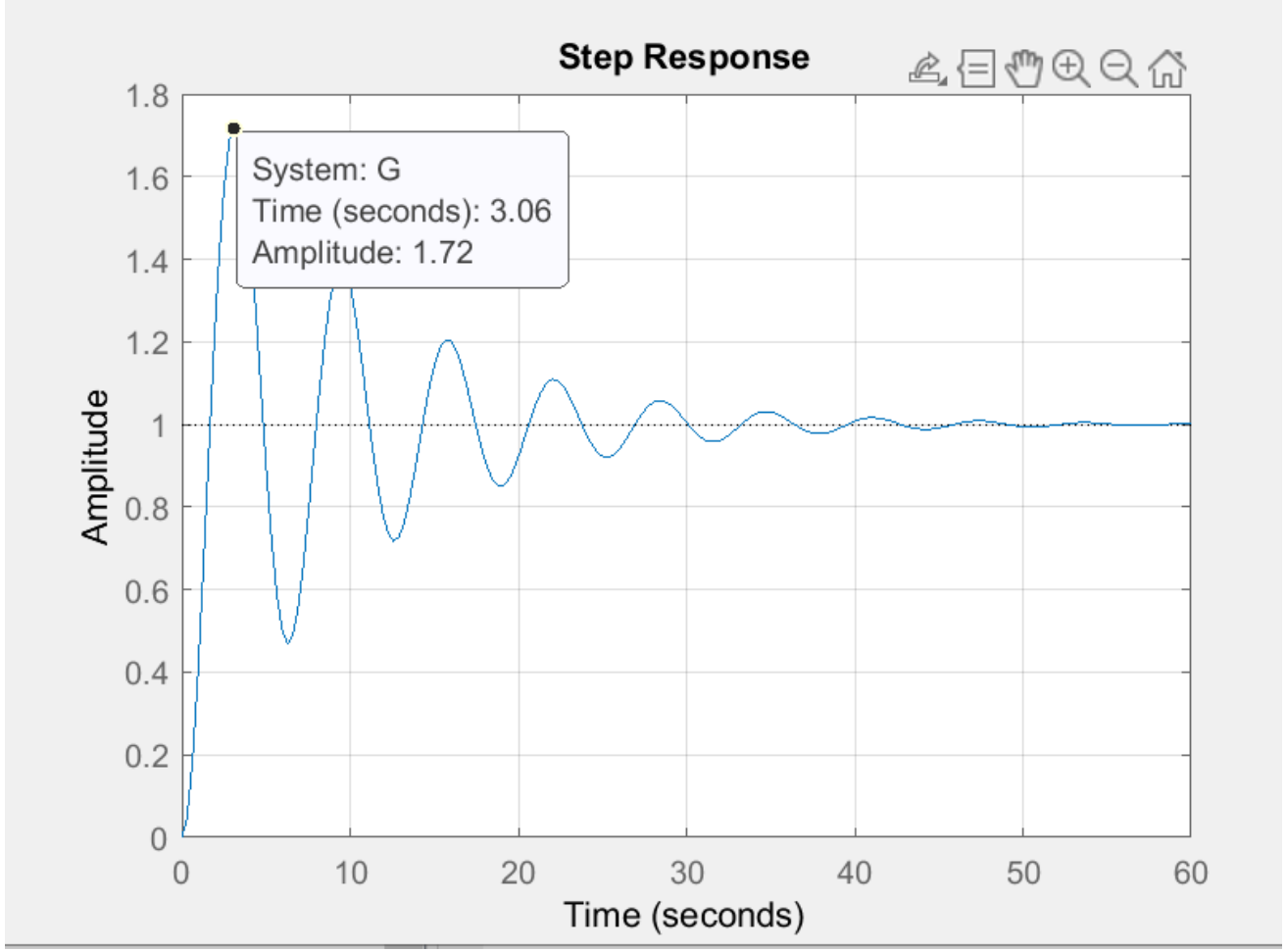
$$G(s) = \frac{X(s)}{U(s)} = \frac{1}{m\,s^2 + bs + k} = \frac{1/m}{s^2 + \frac{b}{m}s + \frac{k}{m}} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Match to 2nd order system:

$$G(s) = \frac{1/m}{s^2 + \frac{b}{m}s + \frac{k}{m}} \quad \Longleftrightarrow \quad G(s) = \frac{\mathrm{K}\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

You can find the damping and stiffness from experimentally derived ω_n and ζ

Finding Model Parameters



Finding the damping ratio, $\zeta \approx 0.1$

Finding the natural frequency $\omega_n \approx 1$

$$G(s) = \frac{X(s)}{U(s)}$$

$$= \frac{\mathrm{K}\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{1}{s^2 + 0.2\,s + 1}$$

$$= \frac{1}{m\,s^2 + bs + k} = \frac{1/\mathrm{m}}{s^2 + \frac{b}{m}s + \frac{k}{m}}$$

Assuming mass m=1

$$b = 2\zeta\omega_n = 0.2$$

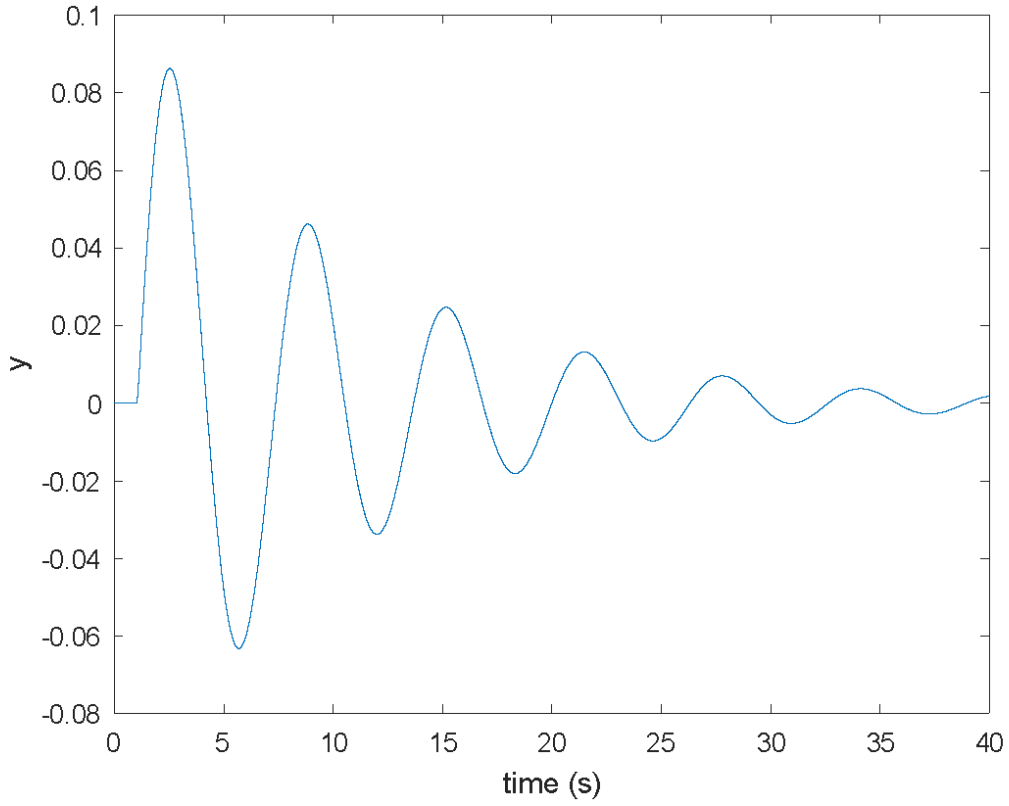
$$k = \omega_n^2 = 1$$

Finding System Model – 2) Impulse response

- System parameters can be found from the response
- To do this experimentally, apply a *small perturbation* and examine the output response

1) Step response

2) Impulse response



Finding Model Parameters

Applying a small perturbation = impulse response: $u = \delta(t)$

$$m\ddot{x} + b\dot{x} + kx = u = \delta(t)$$

Transfer function:

$$m\left(s^2X(s) - sx(0^-) - \dot{x}(0^-)\right) + b(sX(s) - x(0^-)) + kX(s) = 1$$

System starts at a zero initial position and velocity:

$$(ms^2 + bs + k)X(s) = 1$$

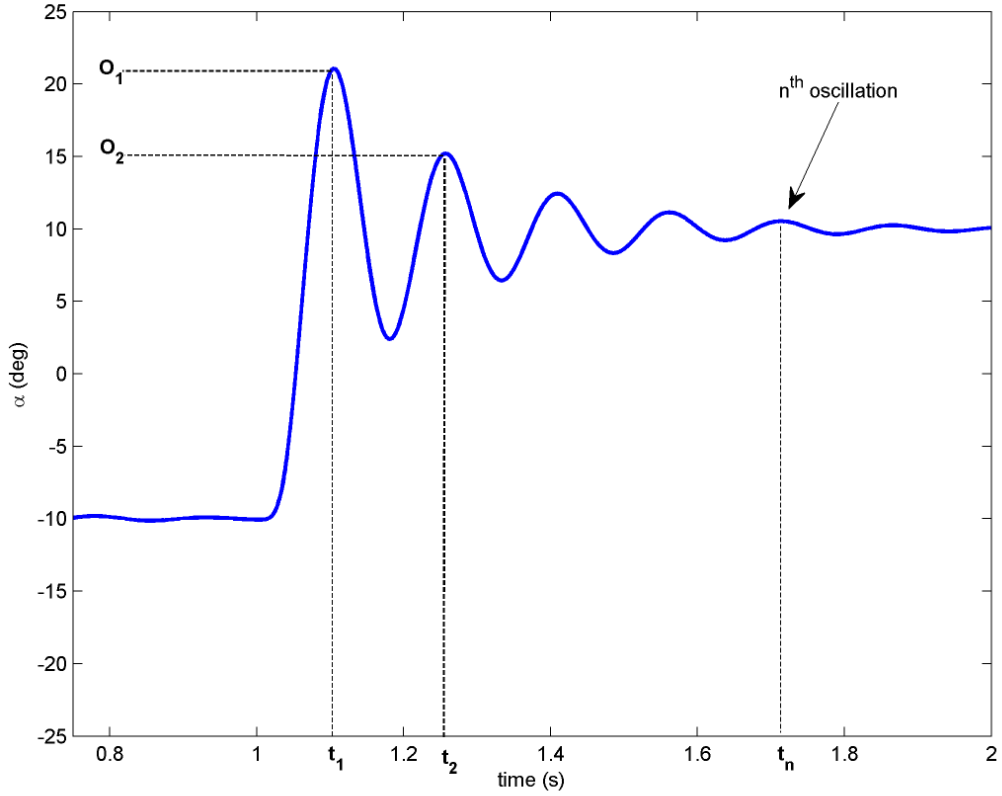
Match to prototype 2nd order system:

$$G(s) = \frac{1/m}{s^2 + \frac{b}{m}s + \frac{k}{m}} \quad \Leftrightarrow \quad G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

You can find the damping and stiffness from experimentally derived ω_n and ζ

2) Impulse response: Finding Natural Frequency and Damping Ratio

Impulse response



Finding the damping ratio, ζ

Logarithmic decrement:

$$\delta = \frac{1}{n}\ln\frac{O_1}{O_n}$$

Damping ratio:

$$\zeta = \frac{\delta}{\sqrt{\delta^2 + (2\pi)^2}}$$

Find the gain

$$\mathrm{K} = \int_0^\infty h(t)\,dt = G(0)$$

Finding the natural frequency, ω_n

Period of oscillations:

$$T_{osc} = \frac{t_n - t_1}{n}$$

Damped natural frequency:

$$\omega_d = \frac{2\pi}{T_{osc}}$$

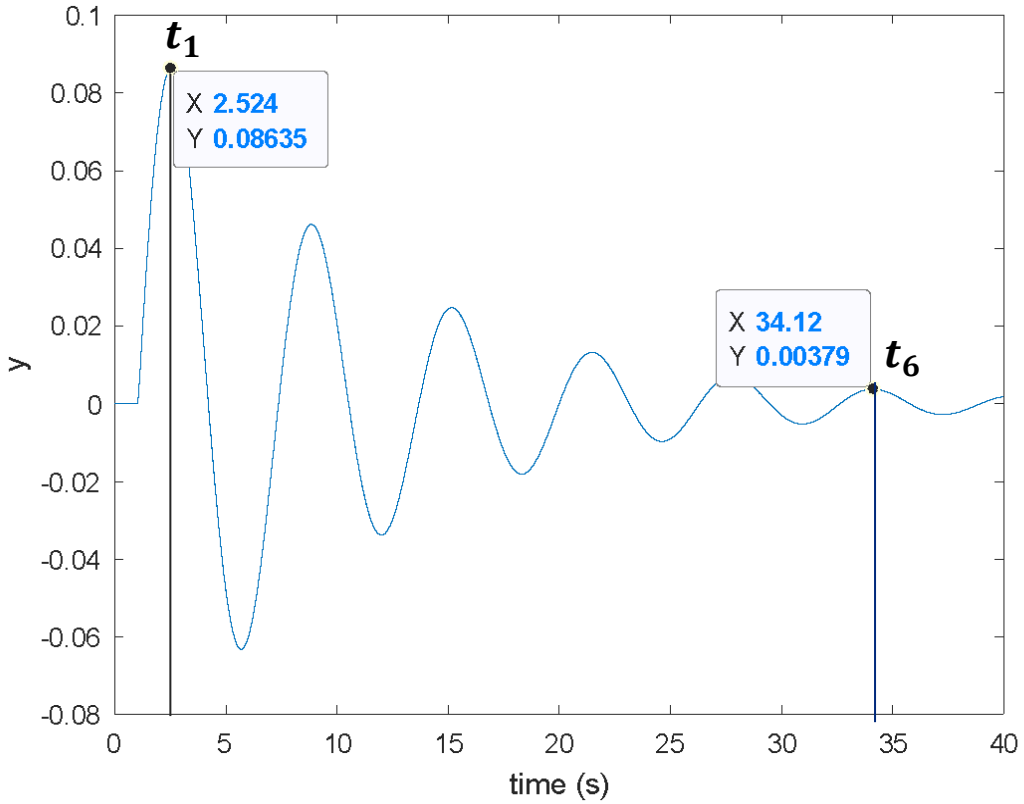
Undamped natural frequency:

$$\omega_n = \frac{\omega_d}{\sqrt{1-\zeta^2}}$$

n *is the number of cycles in the period* $[t_1\ t_n\ ]$

Example: Impulse Response of Mass-Spring-Damper

Response from a small perturbation (i.e. impulse) of mass-spring-damper with mass m=1 kg.



Impulse response

$n = 5$ *is the number of cycles in the period* $[t_1\ t_6\ ]$

$$T_{osc} = \frac{t_6 - t_1}{n} = \frac{34.1 - 2.52}{5} = 6.32$$

$$\omega_d = \frac{2\pi}{T_{osc}} = 0.995$$

$$\delta = \frac{1}{n} ln\ \frac{O_1}{O_6} = \frac{1}{5} ln\ \frac{0.08635}{0.00379} = 0.625$$

$$\zeta = \frac{\delta}{\sqrt{\delta^2 + (2\pi)^2}} = \frac{0.625}{\sqrt{0.625^2 + (2\pi)^2}} = 0.0990 \approx 0.1$$

$$\omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}} = \frac{0.994}{\sqrt{1 - 0.0991^2}} = 1.00$$

Find Stiffness and Damping

Based on the response measurements, calculate the parameters:

$$G(s) = \frac{1}{m\,s^2 + bs + k} = \frac{1/\mathrm{m}}{s^2 + \frac{b}{m}s + \frac{k}{m}} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n\, s + \omega_n^2}$$

Stiffness and damping from (ω_n, ζ):

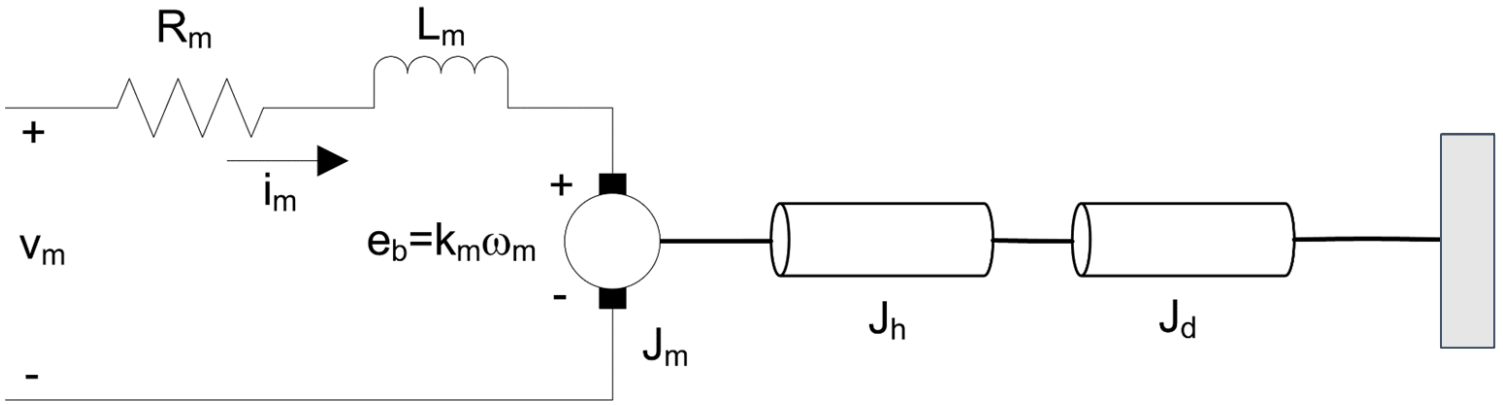
$$\boxed{b = 2\zeta\omega_n m} = 2 * 0.1 * 1 * 1 = 0.2$$

$$\boxed{k = m\omega_n^2} = 1 * 1^2 = 1$$

Given system mass m=1, and the measured ω_n=1 and ζ=0.1 , the damping and stiffness of our system is:

$$b = 0.2$$
$$k = 1$$

Example: Motor Parameter Identification (System identification)



Get voltage-current equation using Kirchoff's Voltage Law (KVL):

$$v_m(t) - R_m i_m(t) - L_m \frac{di_m(t)}{dt} - k_m \omega_m(t) = 0$$

$$v_m(t) - R_m i_m(t) - k_m \omega_m(t) = 0$$

Finding the Resistance and Back-EMF (electro-motive force) Parameters

1. Find resistance:
Stall the motor and measure the current when different voltages are applied

$$v_m(t) - R_m i_m(t) - k_m \omega_m(t) = 0 \quad (\omega_m(t) = 0) \Rightarrow \boxed{R_m = \frac{v_m}{i_m}}$$

2. Finding back-emf k_m:
With the motor free spinning, apply different voltage and measure the corresponding current and speed:

$$v_m(t) - R_m i_m(t) - k_m \omega_m(t) = 0 \Rightarrow \boxed{k_m = \frac{v_m - R_m i_m}{\omega_m}}$$

Basic Procedure:

1.Apply a range of voltages (e.g. -8V, -7V, … 8V) and find R_m and k_m for each.

2.Take average of result.

Example Result: Resistance

Applied Voltage (V)	Measured Current (A)	Resistance (Ω)
-3	-0.31	9.7
-2	-0.19	10.5
-1	-0.19	10.4
1	0.95	10.5
2	0.19	10.5
3	0.30	10.0
	Average R_m =	10.3

Example Result: Back-EMF

Applied Voltage (V)	Measured Speed (rad/s)	Measured Current (A)	Back-EMF (V-s/rad)
-3	-68.8	-0.00161	0.0434
-2	-44.3	-0.00403	0.0442
-1	-22.3	-0.00202	0.0439
1	22.1	0.00282	0.044
2	44.4	0.00524	0.0439
3	69.0	0.00363	0.043
		Average k_m =	0.0437

Experimental modeling using Frequency Response

- Using asymptotical bode plots to find the transfer function
- Review of frequency responses: see additional file Lecture_Frequency_Response.pdf

Bode plots

Bode Plots: 2 separate logarithmic plots

- Magnitude plot: Amplitude in decibels (dB), where $dB = 20 \log_{10} |G(j\omega)|$ vs. frequency $\log_{10} \omega$
- Phase plot: Phase angle in degrees, $\angle G(j\omega)$ vs. frequency $\log_{10} \omega$

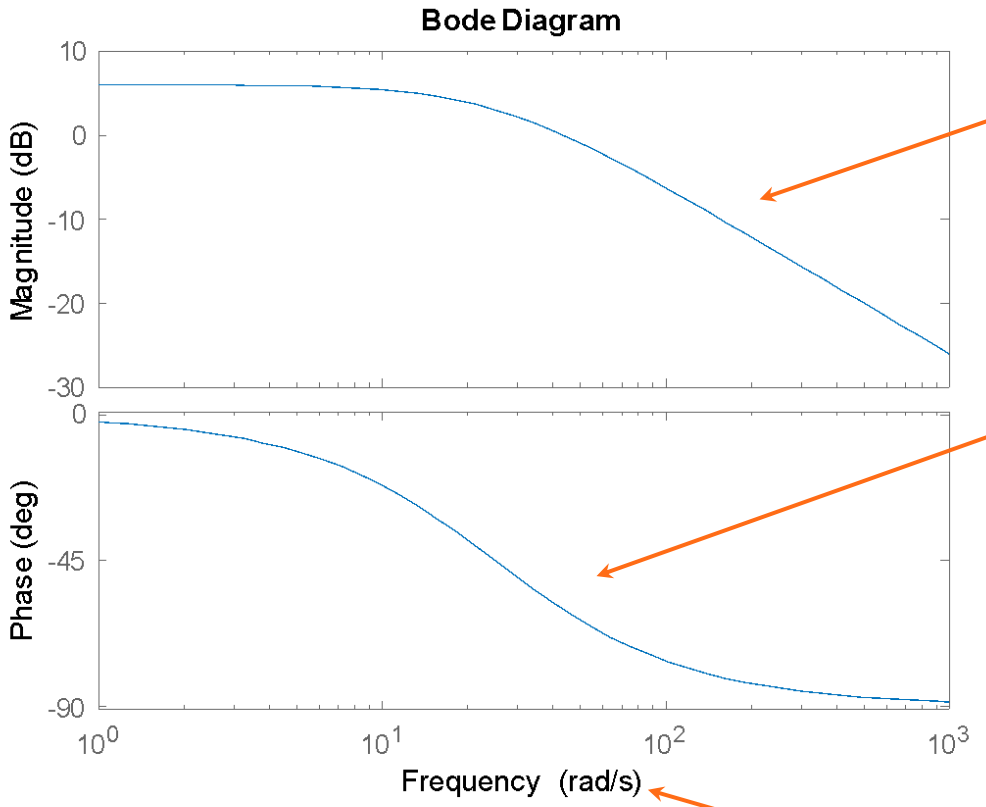
Bode plots

2 separate logarithmic plots

- Magnitude plot: Amplitude in 20 log10 |G(jω)| decibels vs. frequency $\log_{10}\omega$
- Phase plot: Phase angle in degrees, ∠G(jω) vs. frequency $\log_{10}\omega$

Decibel (dB)
$\mathbf{dB} = 20\log_{10}|H(i\omega)|$ is a logarithmic unit, used to express the ratio between two values of a physical quantity.

Decade ($\log_{10}\omega$): is a unit for measuring frequency ratios on a logarithmic scale, with one decade corresponding to a ratio of 10 between two frequencies.



Plot gain of system in dB vs. frequency
$|H(j\omega)|_{dB} = 20\log_{10}|H(j\omega)|$

Plot phase φ vs. frequency

$\log_{10}\omega$

The reason why Bode plots have logarithmic scale

Consider the following transfer function:

$$G(s) = \frac{K(s+z_1)(s+z_2)\dots(s+z_k)}{s^m(s+p_1)(s+p_2)\dots(s+p_n)}$$

The frequency response:

$$G(j\omega) = \left.\frac{K(s+z_1)(s+z_2)\dots(s+z_k)}{s^m(s+p_1)(s+p_2)\dots(s+p_n)}\right|_{s\to j\omega}$$

The main reason why Bode plots are logarithmic:

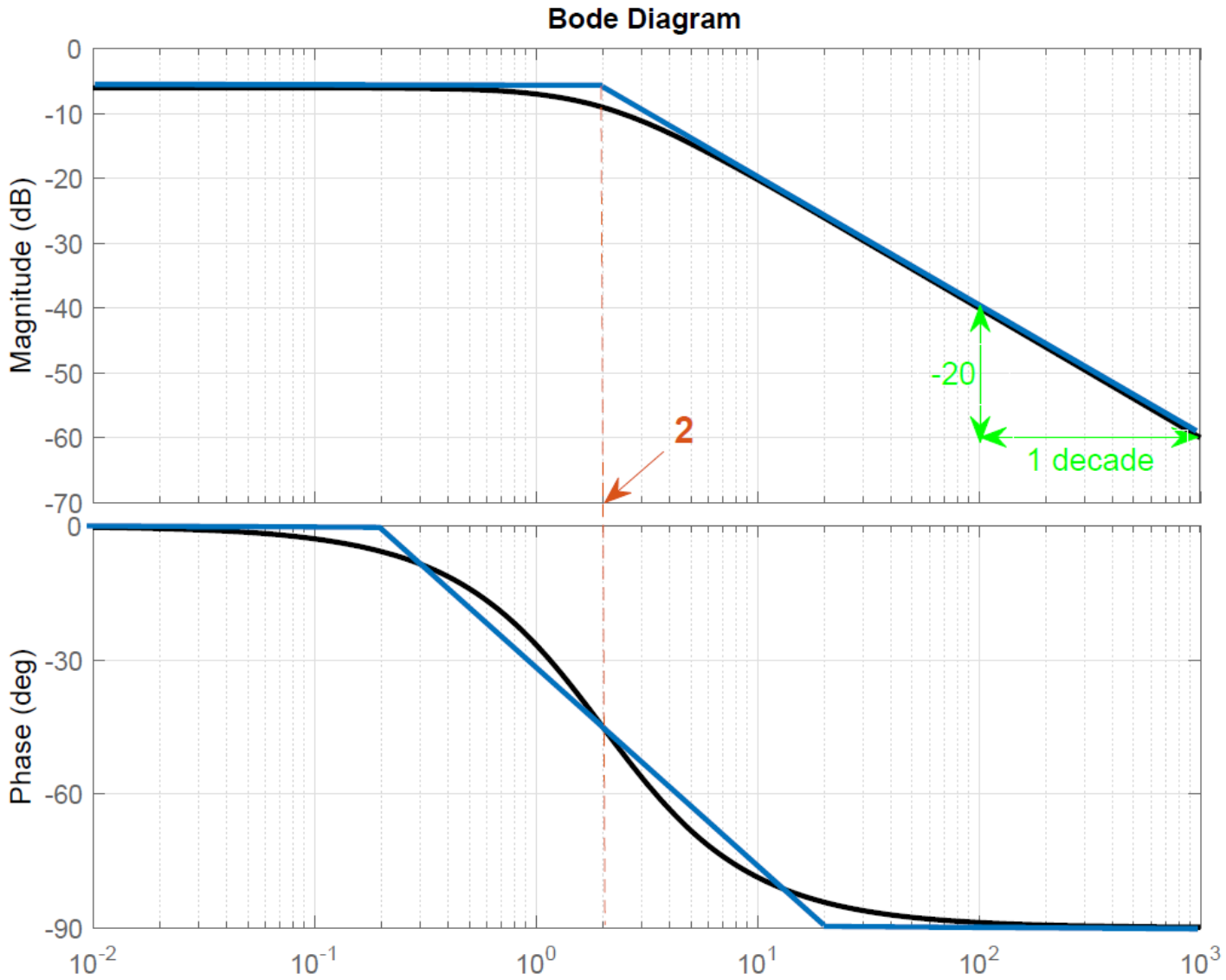
$$\begin{aligned}
20\log|G(j\omega)| =& 20\log K + 20\log|(j\omega+z_1)| + 20\log|(j\omega+z_2)| + \dots \\
&- 20\log|(j\omega)^m| - 20\log|(j\omega+p_1)| - \dots - 20\log|(j\omega+p_n)| \\
\angle G(j\omega) =& \angle(j\omega+z_1) + \angle(j\omega+z_2) + \dots + \angle(j\omega+z_k) \\
&- m\times 90^o - \angle(j\omega+p_1) - \dots - \angle(j\omega+p_n)
\end{aligned}$$

- Magnitude response: The algebraic sum of Log magnitude response of each term would yield the total magnitude response in dB.
- Phase response: The sum of phase response of each term yields the total phase response in degrees.

Logarithmic scale, Decibel (dB) and Decade

- A logarithmic scale is a nonlinear scale used when there is a large range of quantities. Common uses include earthquake strength, sound loudness, light intensity, and pH of solutions.
- The decibel (dB) is a logarithmic unit used to express the ratio between two values of a physical quantity.
- One decade is a unit for measuring frequency ratios on a logarithmic scale, with one decade corresponding to a ratio of 10 between two frequencies. It is especially useful when referring to when describing frequency response of electronic systems, such as audio amplifiers and filters.

Example: Bode plots of $G(s) = \frac{1}{(s+2)}$ - Straight-lines approximation



Experimental modeling using Frequency Response

Using asymptotical bode plots to estimate the transfer function

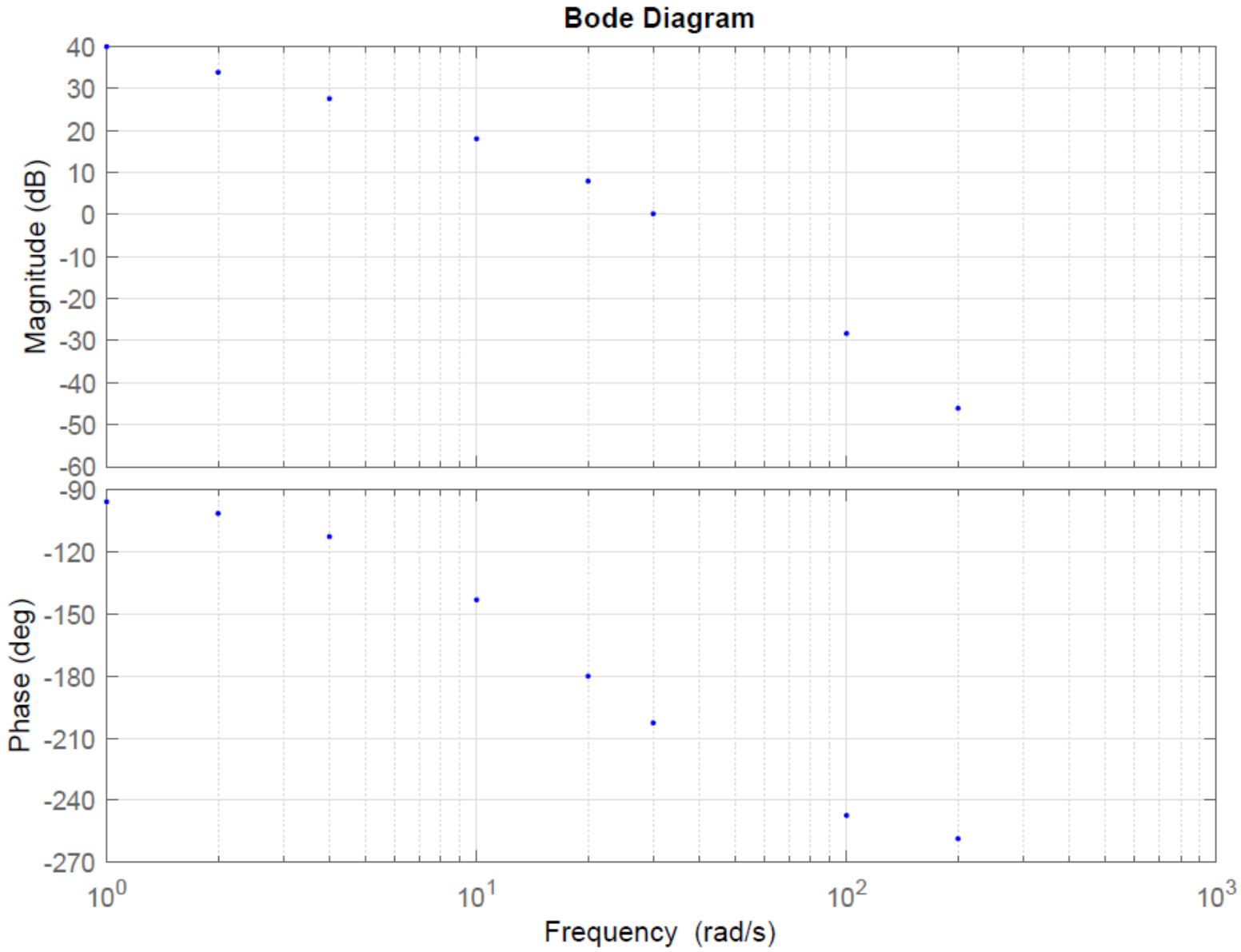
- Plot amplitude and phase data on the Bode Plot
- Fit straight-lines to amplitude data.
- Use relationships to determine the transfer function

- Slope must be 0 dB/decade, ± 20 dB/decade, ± 40 dB/decade, etc.
- Straight-lines must pass through the data points at the lowest frequency and highest frequency.
- Straight-lines should be close to the data points.

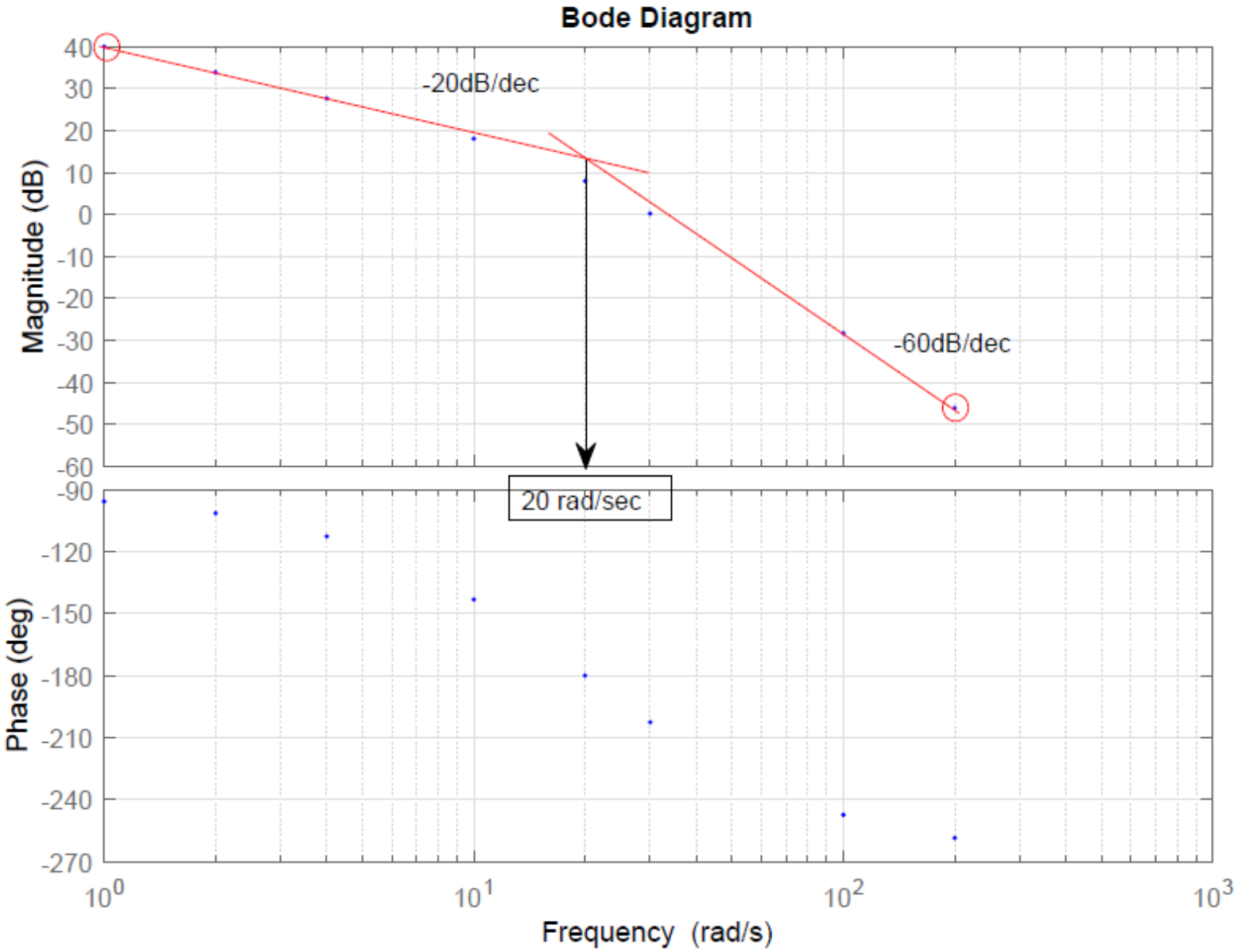
Experimental modeling using Frequency Response

- Check asymptotic magnitude plot and phase plot in low frequency
  - if the initial magnitude plot with slope 0 dB/dec and the initial phase at $0°$
    $\Rightarrow$ the transfer function is type zero system (no "s" in the system).
  - else if the initial magnitude plot with slope -20 dB/dec and the initial phase at $-90°$
    $\Rightarrow$ the transfer function has $\frac{1}{s}$.
  - else if the initial magnitude plot with slope -40 dB/dec and the initial phase at $-180°$
    $\Rightarrow$ the transfer function has $\frac{1}{s^2}$.
  - else if the initial magnitude plot with slope $+20$ dB/dec and the initial phase at $+90°$
    $\Rightarrow$ the transfer function has s.
  - else if ....
- Check the following slope in the magnitude plot and phase shift in phase plot, then find each break frequency a (Intersection of two lines), from the lowest to the highest,
  - If the magnitude slope change by $+20$ dB/dec and the phase shift is $+90$ degree,
    $\Rightarrow$ then the transfer function has $(s+a)$.
  - Elseif the magnitude slope change by -20 dB/dec and the phase shift is -90 degree,
    $\Rightarrow$ then the transfer function has $\frac{1}{(s+a)}$.
  - Elseif the magnitude slope change by $+40$ dB/dec and the phase shift is $+180$ degree,
    $\Rightarrow$ then the transfer function has $(s+a)^2$.
  - Elseif the magnitude slope change by -40 dB/dec and the phase shift is -180 degree,
    $\Rightarrow$ then the transfer function has $\frac{1}{(s+a)^2}$.
  - ....

Example: Find the transfer function using frequency data



Example: Find the transfer function using data



Low frequency slope and phase

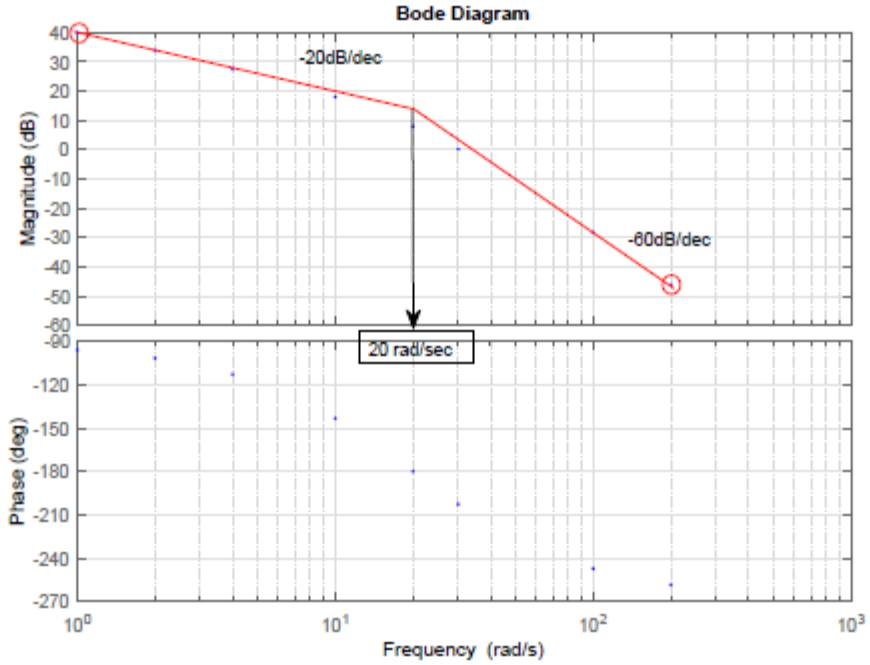
- slope$= -20$ dB/decade and phase $= -90^o$

$\Rightarrow$ Indicates a integrator term $\frac{1}{s}$

High frequency slope and Phase

- slope$= -60$ dB/decade and phase $= -270^o$

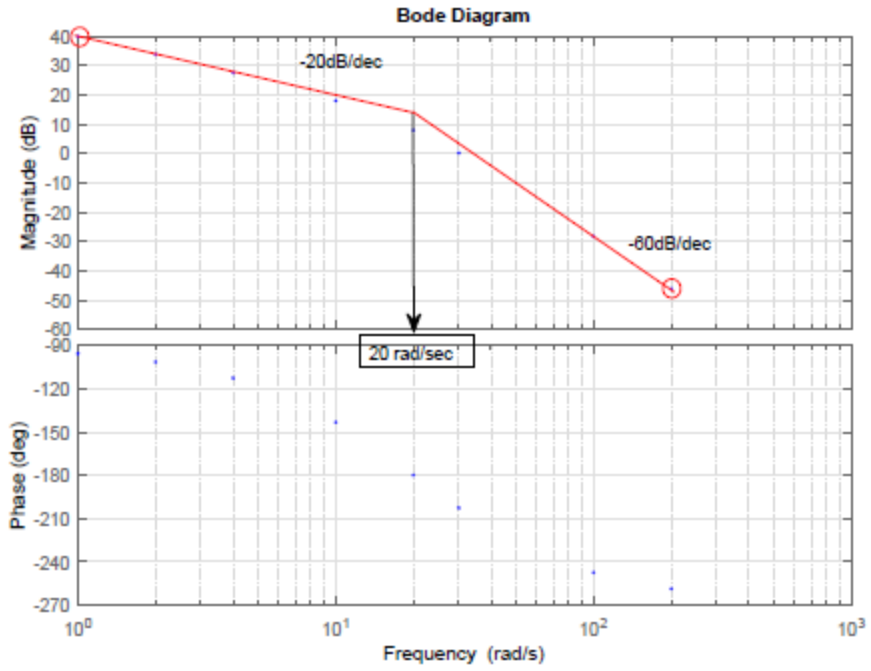
Indicates a 3-order system



Break frequency

- Only one internal break point at intersection of low frequency and high frequency asymptotes
- slope change $= (-60) - (-20) = -40$ dB/decade and phase shift $= (-270) - (-90) = -180^o$
- break point frequency $= 20 \ \ \mathrm{rad/s}$

Indicates 2nd-order pole at break frequency: $\frac{1}{(s+20)^2}$ (two same poles)



System transfer function:

$$\text{TF} = \frac{K}{s(s+20)^2}$$

Calculate Gain K

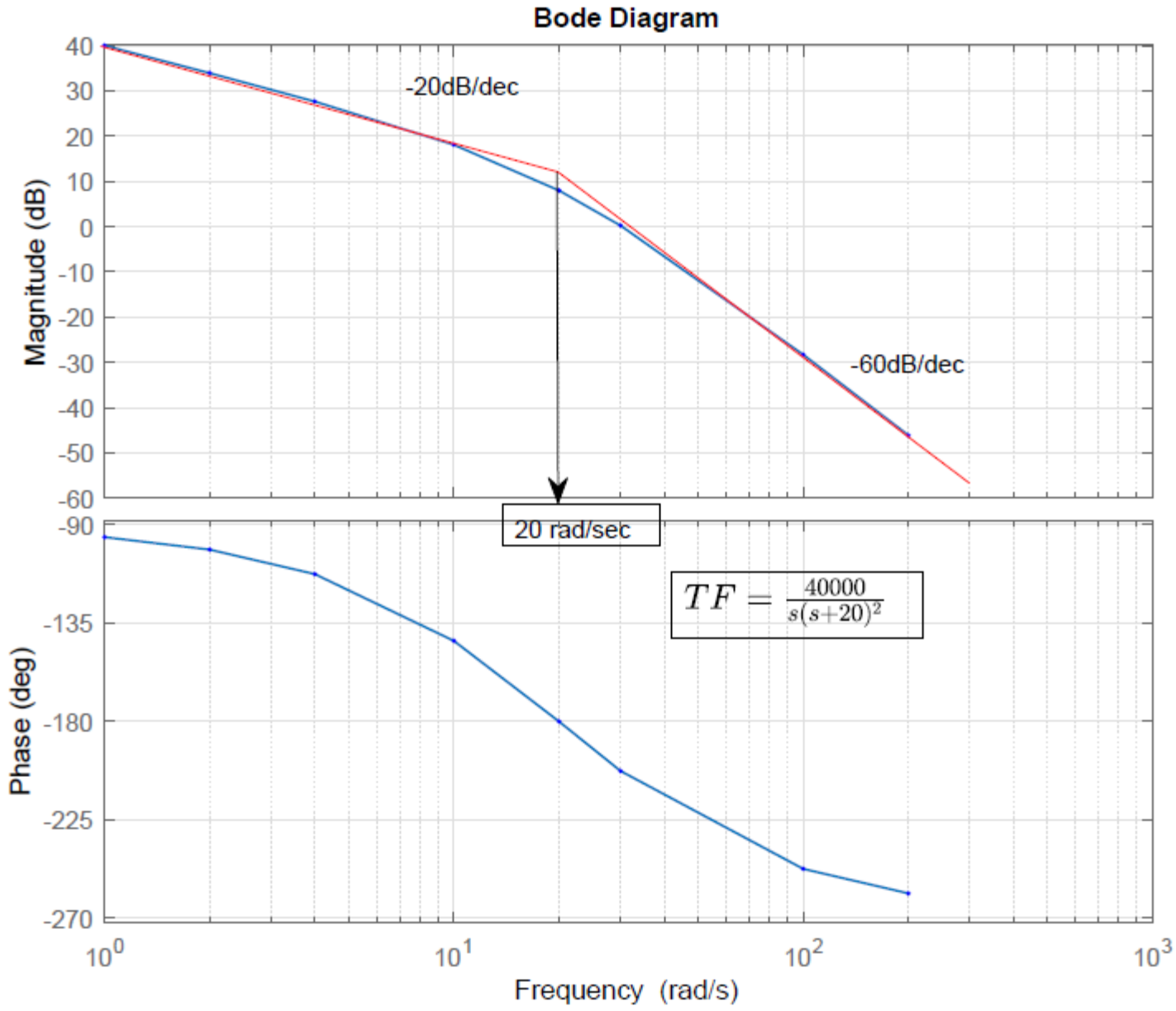
- Find a point at low frequency $\omega \ll 20$: At $\omega = 1$ rad/s, Amplitude=40 dB
- Gain: If $\omega \ll 20$: $20\log|\frac{K}{j\omega_{(j\omega+20)^2}}| \approx 20\log|\frac{K}{(\omega)(20)^2}|$

$\omega = 1$, $20\log|\frac{K}{j(j+20)^2}| \approx 20\log|\frac{K}{20^2}| = 20\log(\frac{K}{400}) = 40$ dB

$\Rightarrow \frac{K}{400} = 10^{(40/20)} = 100 \Rightarrow K = 40000.$

$$\text{TF} = \frac{40000}{s(s+20)^2}$$

Asymptotic and Actual bode plots



Experimental Modelling: Takeaways

Benefits

- Don't require system specifications/parameters (if they are not available)
- Quickly derive low-order models
- Fairly accurate (but only for certain input range)

Drawbacks

- Cannot be used for unstable or dangerous systems, e.g. cart on track would run off track
- Model derived may only be accurate for certain input signal range